

COURSE STRUCTURE

Course Code	BCA20110				
Course Category	Program Major				
Course Title	Programming in JAVA				
Teaching Scheme and Credits	L	T	Laboratory/Practical	Project	Total
Weekly load hrs.	3	-	2	-	5
Credits	3	-	1	-	4
Assessment Schema Code	TL3				
<u>Pre-requisites:</u> Need basics of Programming and Object-Oriented Programming concepts.					
Course Objectives: Knowledge: To understand Java programming concepts with respect to i) Object Oriented Programming ii) Data Structures in Java Programming Skills: Design skills of i) Desktop Applications Development ii) Object oriented programming Attitude: To develop following i) Programming Skills ii) Build Stand-alone Application in Complex situation					
<u>Course Outcomes:</u> a) Identify classes, objects, class members and relationships for a given problem. b) Design end to end applications using object-oriented constructs. c) Apply collection classes for storing java objects. d) Use Java APIs for program development. e) Handle abnormal termination of a program using exception handling.					
Course Contents Unit 1: Introducing to Java:4 A Short History of Java, Features of Java, Java Environment – Compiler, Interpreter, JVM, Simple java, program, Types of Comments, Declaring single and multi-dimensional arrays, Accepting input using Command line arguments, Accepting input from console (Using BufferedReader and Scanner class) Unit 2: Classes and Objects:8 JSP Basics, JSP Derivatives, Implicit Object in JSP, JSP Actions, JSP Database Handling, Session, Handling in JSP, JSP Beans, JSP Custom Tag Packages -Creating, Accessing and using Packages Unit 3: Inheritance:10 Inheritance Basics (extends Keyword) and Types of Inheritance, Superclass, Subclass and use of Super Keyword, Method Overriding and runtime polymorphism. Use of final keyword related to					

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variable, method and class, Use of abstract class and abstract methods

Interface and Packages-Defining and Implementing Interfaces, Runtime polymorphism using interface

Unit 4: Collections -8

Wrapper Classes, Introduction to the Collection framework, List – ArrayList, LinkedList and Vector,

Set - HashSet, TreeSet, and LinkedHashSet, Map – HashMap, LinkedHashMap, Hashtable and TreeMap

Interfaces such as Iterators, ListIterators, Enumerations

Unit 5: Exception Handling :4

Exception class, Checked and Unchecked exception, Catching exception and exception handling – try, catch, finally, throw and throws, multiple catch block, Creating user defined exception

Unit 6: I/O:3

String class(basic methods) , String Buffer class File class DataInputStream and DataOutputStream class

Unit 7:Swing:8

What is Swing? The MVC Architecture and Swing, Layout Manager and Layouts, The JComponent class Components – JLabel, JButton, JText, JTextArea, JCheckBox, JRadioButton, JList, JComboBox, JMenu and JPopupMenu Class, JMenuItem, Dialogs (Message, confirmation, input), JFileChooser, Event Handling: Event sources, Listeners – ActionListener, ItemListener, Mouse and Keyboard Event Handling, Adapters – MouseAdapter, KeyAdapter, Anonymous inner class

Learning Resources:

Reference Books:

1. Java 2 programming black books, Steven Horlizer

Supplementary Reading:

2. Complete reference Java by Herbert Schildt(5th edition)

3. Java Complete Reference Patric Naughton, Herbert Schildt, TMH,7th Ed.

4. Complete Reference- J2EE Jim Keogh, TMH.

5. Inside Servlets- Dustine R Callway

7. Programming with Java, A primer, Fourth edition, By E. Balagurusamy

8. Core Java Volume-I-Fundamentals, Eighth Edition, Cay S. Horstmann, Gary Cornell, Additional Reading

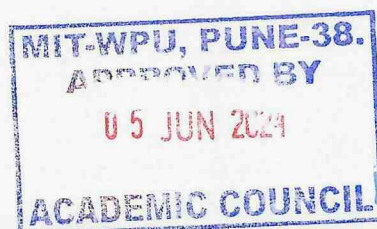
Websites : <https://www.tutorialspoint.com>

<https://www.javatpoint.com>

MOOCS : Spoken Tutorial, Khan Academy

Pedagogy:

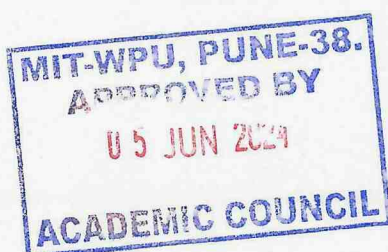
- Practical learning
- project handling
- problem solving assignment
- conceptual and contextual learning
- PPT Presentation & Interactive discussion



Laboratory Experiments / Software based Practical

Sr No.	Contents
1	Implementation of basic concepts of Java, constants, operators, decisions and loops etc.
2	Implementation of Classes, Objects, Creation of objects in various ways, constructors, Packages
3	Implementation of Inheritance, types of inheritance, aggregations, Interface, Polymorphism and Packages
4	To study Collection framework.
5	To study Exception, types of exceptions, throw, throws.
6	I/O Handling-String and StringBuffer class, File class, DataInputStream and DataOutputStream class
7	To Study GUI development using SWING

M. Bechu



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